

Convolutions

Time invariance:

$$y(t) = \sin(t)x(t) \quad (1)$$

is not time invariant. If $x(t) \rightarrow x(t - t_0)$, then we would have

$$y(t) = \sin(t)x(t - t_0) \quad (2)$$

however,

$$y(t - t_0) = \sin(t - t_0)x(t - t_0) \quad (3)$$

which is not the same.

We define a convolution

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] = y[n] \quad (4)$$

In continuous time

$$x * h = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \quad (5)$$